



Introduction To Programming Using C++ Level 2 Function Overloading



Introduction To Programming Using C++

Level 2

Function Overloading

Definition

- Function Overloading in C++ means having multiple Functions with the same name but different Parameters.
- Functions can have the same name but differ in:
 - The number of Parameters.
 - The types of Parameters.
- Function Overloading is a form of Polymorphism because the same Function name can represent different implementations depending on the Parameters used.

Explanation

- The compiler determines which Function to call based on the number and types of the Arguments provided.
- The following differences are not enough for Function Overloading:
 - ▨ Different Parameter names only.
 - ▨ Different Return Types only.
- Functions with the same name and exactly the same Parameters cannot be overloaded because the compiler cannot distinguish between them.
- C++ supports Function Overloading, which allows us to use the same Function name for related operations.

Procedures

1. Create multiple Functions with the same name.
2. Change the number or types of their Parameters.
3. Call the Function with the required Arguments.
4. The compiler selects the matching Function based on the Arguments.

💡 Example

```
double MySum(double a, double b)
{return (a + b);}
int MySum(int a, int b)
{ return (a + b);}
int MySum(int a, int b, int c)
{ return (a + b + c); }
int MySum(int a, int b, int c, int d)
{ return (a + b + c + d);}
```

- ⚡ All Functions have the same name:
MySum
- ⚡ But they have different Parameter lists.

➡ Result

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Note

- ◆ Function Overloading allows us to use one meaningful name for several related Functions.
- ◆ C++ also has Built-in overloaded Functions, such as `max()`, which can work with different types.

⊘ Warning

- ◆ Changing only the Parameter names does not create an overload.

```
int MySum(int a, int b);
```

```
int MySum(int x, int y); // ✕ Conflict
```

- ◆ Changing only the Return Type does not create an overload.

```
int MySum(int a, int b);
```

```
double MySum(int a, int b); // ✕ Conflict
```



Comparison

▨ Function Overloading

- ✓ Same Function name.
- ✓ Different number or types of Parameters.
- ✓ The compiler selects the appropriate Function based on the Arguments.
- ✓ Makes related operations easier to read and use.

▤ No Function Overloading

- ✓ Each Function needs a different name.
- ✓ Similar operations may require several different Function names.
- ✓ The code can become less consistent when many related operations exist.

✦✦ Summary

- ✓ Function Overloading means using the same Function name with different Parameters.
- ✓ Functions can differ by the number or types of Parameters.
- ✓ Return Type alone cannot be used for Overloading.
- ✓ Parameter names alone cannot be used for Overloading.
- ✓ C++ supports Function Overloading.
- ✓ Function Overloading is a form of Polymorphism.

Thank you 😊